EE5203 L12 Worksheet

System integration and diagnosis - Basri Erdogan

Expected time: 30-45 minutes

Policy: Attempt every question before consulting the worked solutions. Several questions ask about **your own project** — answer for it, not in general.

Part A - Resource conflicts

1. Name the three species of failure and, for each, the tool that finds it.
2. Here is a proposed resource map. Find **three** conflicts and explain each.

Resource	Pin(s)	Owner
TIM2 CH1	PA5	LED PWM, 1 kHz
TIM2 CH2	PA1	servo PWM, 50 Hz
ADC1 IN1	PA1	light sensor
SPI1 SCK	PA5	SD card
USART2	PA2, PA3	telemetry
TIM4	PB6	input capture

3. Build the resource map for **your own project**. One row per resource, one owner each, and mark anything you are unsure about.
4. Explain why `HAL_Delay` inside an interrupt callback can hang forever. Which peripheral does it depend on, and what is the priority condition?

Part B - Timing budgets

5. Write the formula for the load of an activity from its cost and rate.
6. Compute each load and the total.

Activity	Cost	Rate	Load %
Tick interrupt	2 µs	1000 Hz	
ADC read (two channels)	40 µs	200 Hz	
UART line, 24 chars @ 115 200		10 Hz	
LCD line, 16 chars @ 100 kHz	7.2 ms	4 Hz	
Total			

7. Which activity dominates that budget, and by what factor over the next largest? What would you change first, and what would it save?
8. A team wants the LCD refreshed at 30 Hz. Compute the load at 100 kHz and at 400 kHz. Is either acceptable? Give a design that achieves a *visibly* responsive display without either cost.
9. Build the timing budget for **your own project**. State the total, name the dominant item, and say what you would cut first if you had to.

Part C - Diagnosis

10. For each symptom, name the most likely species and the first thing you would check.

Symptom	Species	First check
Display garbles only while the servo sweeps		
Telemetry stops permanently after ~10 minutes		
The servo never moves, from the very first run		
The board resets every few minutes under load		
Readings are correct but arrive in bursts		

11. You have six subsystems and one intermittent fault. Describe a bisection plan, and state the worst-case number of

steps. Show the reasoning.

12. Give three instruments for observing a system you cannot stop, with the approximate cost and what each reveals.
13. Turn this into a usable bug report: *"it hangs sometimes."* State what you would measure and what a good report would say.
14. Explain how `git bisect` finds the commit that introduced a fault, and what you must be able to do at each step for it to work.

Part D - Requirements and evidence

15. Rewrite each as a testable requirement.

Vague	Testable
"The logging is reliable"	
"The motor responds quickly"	
"The sensor is accurate"	
"The system is robust"	

16. Write three testable requirements for **your own project**, each with the test procedure and the result you would accept as a pass.
17. Why does an empty known-issues list read badly at a readiness review? What does a good one look like?

Part E - The physical layer

18. Explain why two supplies without a common ground produce failures that are hard to diagnose. What is the symptom?
19. A servo stalls and the board resets. Explain the mechanism, and give two fixes — one electrical, one in software.
20. Give two symptoms that a loose jumper and a software race have in common, and one test that distinguishes them.

Exit self-assessment

☐ I have a complete resource map for my project with no unresolved conflicts.
☐ I have a timing budget with a stated total.
☐ I can name the species of a failure from its symptom.
☐ I can describe a bisection I performed.
☐ My requirements are testable and I know how each is tested.